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# STRUCTURED SLOT MEMORY FOR MASKED-EVIDENCE WORLD-MODEL INTERFACES

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## ABSTRACT

Finite-context world models can observe a decisive visual cue and later act as if it was never seen. We treat this as an interface problem rather than an architecture label: old evidence must survive beyond the legal context window and must also be exposed through a frozen host representation in a form a later consumer can use. A valid memory claim must therefore separate four questions: whether the cue lies outside the legal window, whether it is retained after the window expires, whether the frozen interface exposes it, and whether a bounded controller can act on it. We instantiate this ladder with a self-supervised masked-evidence JEPA objective and a compact slot-memory writer. Training never sees cue labels, cue crops, or key-frame annotations; salient patch sets are mined from the rendered stream, withheld from the visible context, and predicted from a causal memory state, while cue labels enter only afterwards for a fail-closed audit under matched full, reset, and no-state conditions. Across a suite of OGBench render decks, controlled evidence ages, and seeds, the slot interface clears every registered retention gate while matched controls stay at chance, and the largest margins over recurrent GRU, LSTM, and Mamba-lite writers appear exactly where evidence is old or spread across object-like structure. Trained on a mixture of delays, the same interface generalizes to unseen delays; the memory is bounded in output throughout, and a streaming variant that evicts old tokens additionally bounds the attention context, scaling to far longer horizons at constant per-step memory and compute. The advantage survives matching the carrier capacity and adding structured state-space, gated-slot, and retrieval memories, does not depend on a colour shortcut when the cue identity is decorrelated from colour, and transfers into a frozen LeWorldModel host, where the injected memory exposes old evidence in the host’s own predictive output beyond its context window. The contribution is a scoped audit-and-method result: persistent old-evidence memory in latent world models is a property of the target–carrier–host–consumer interface, not a universal property of any single recurrent or state-space module.

## 1 INTRODUCTION

Finite-context world models often fail in a way that is easy to miss: they can observe a decisive cue early and later produce a representation that behaves as if the cue was never present. This matters for physical intelligence because action-relevant evidence is rarely synchronized with the moment of action. A robot can read a sign before turning away, see a door state before entering a hallway, or observe an object identity before that object becomes occluded. If the interface to a frozen host exposes only a recent window, then locally plausible prediction is not enough; the old evidence must survive long enough, in a usable form, to change a later decision.

We study this failure as an *interface* problem rather than a raw memory capacity problem. The question is not only whether a hidden state can memorize something. The stricter question is whether old visual evidence, after it has left the legal context window, is (i) *retained* in a causal memory state, (ii) *exposed* through a host-facing representation, and (iii) *usable* by a bounded downstream controller. These three claims are routinely conflated. A model may store old evidence internally yet fail to expose it in its output. A linear readout may decode a label through a shortcut if the current endpoint still leaks the cue. A native controller may succeed only because it received the true goal

rather than the memory-selected one. Our evaluation is designed to close each of these loopholes with matched controls.

We introduce a *masked-evidence slot-memory interface* (Figure 1). Training is self-supervised: it never receives cue labels, cue crops, or manually selected key frames. A target miner divides the causal render stream into temporal bins, scores visual saliency, selects salient patch sets, and removes them from the visible stream. A compact slot transformer must then predict the stop-gradient latent targets of those hidden patches from a small persistent memory state. Cue labels enter only *after* training, when a post-hoc ridge readout tests whether the learned memory exposes the old cue under matched full, reset, and no-state conditions.

The central empirical point is structural rather than merely numerical. Under the same caches, target mining, patch target encoder, loss, optimizer, and readout, replacing the slot writer with GRU, LSTM, or Mamba-lite recurrence weakens long-delay exposure. The recurrent baselines often solve short, waypoint-like rows, which is useful evidence that the task is not impossible; their failure appears when the cue is old or distributed across object-like visual evidence. Slots help because the target itself is a *set* of visual facts, and a memory set is a better carrier for that target than a single recurrent vector.

Concretely, on twelve OGBench render decks across a sweep of controlled evidence delays and three seeds, the slot writer passes every registered retention cell (36/36 environment-age rows, 108/108 seed cells) with mean full-memory balanced accuracy 0.983 while reset and no-state controls stay at 0.250 and 0.248. The strongest recurrent baseline reaches 0.803 mean full accuracy and 22/36 rows, and its margin behind the slot writer widens as evidence ages. Where audited fixed controllers exist, memory-selected targets execute at 0.957 mean success across 27/36 rows.

## Contributions.

- A **fail-closed audit** for old-evidence memory in frozen world-model interfaces, separating retention, host exposure, and bounded use with matched reset / no-state controls.
- A **label-free masked patch-set JEPA objective** that trains a slot memory writer from automatically mined salient patches, with full pseudocode (Algorithms 1–2).
- A **controlled architecture comparison** across twelve OGBench decks, three evidence ages, and three seeds, isolating the writer format while holding every other component fixed, extended with capacity-matched and structured-memory baselines (SlotSSM-style, gated slot attention, top- $k$  retrieval, cached attention).
- A **bounded streaming formulation**  $M_t = f_\theta(M_{t-1}, Z_{t-K+1:t})$  with token eviction and constant per-step memory/compute, plus a **delay-generalization** and **age-scaling** study to age 128.
- A **frozen-LeWM integration** showing the same interface exposes old evidence in the host’s own output beyond its context window, and a **shortcut-broken benchmark** confirming retention of object identity rather than colour.

Our contribution is best read as a *fail-closed evaluation protocol plus a controlled demonstration that target-carrier structural alignment improves host-visible old-evidence exposure*, rather than a wholly new slot-memory architecture: slot memory (Jiang et al., 2024; Zhang et al., 2024; Anonymous, 2026), masked JEPA prediction (Anonymous, 2026), and memory-augmented world models (Anonymous, 2025; LeWorldModel Team, 2026) each exist; the novelty is the interface audit that separates storage, host exposure, and use and shows where structural alignment matters.

## 2 RELATED WORK

**World models under memory pressure.** Latent world models compress observations into predictive state for control (Hafner et al., 2023). Recall-to-Imagine makes long-term memory and credit assignment explicit bottlenecks for model-based RL (Samsami et al., 2024). Diffusion and video world models face a related pressure: short observation windows can produce plausible futures that drift from earlier evidence. LoopNav turns this into looped spatial-consistency tests (Lian et al., 2025); StateSpaceDiffuser, long-context state-space video models, and memory-expert diffusion world models add state-space or expert memory paths to preserve temporal consistency across long

rollouts (Savov et al., 2025; Po et al., 2025; Stapf et al., 2026). We ask a narrower but stricter interface question: not whether a generator looks temporally consistent, but whether old evidence that is illegal for the current context is still exposed in a frozen host-facing representation.

**Partial observability and memory in RL.** Partially observable RL has long treated memory as a substitute for unavailable Markov state. PO-Dreamer adds memory guidance to world models under partial observability (Li et al., 2025), and memory traces replace fixed observation windows with compact, analyzable history summaries (Eberhard et al., 2025). These works optimize agents or value estimates. In contrast, our audit isolates a representation-interface claim: the downstream readout receives the same endpoint under full, reset, and no-state conditions, so success must come from retained old evidence rather than a policy learning to exploit a different observation stream.

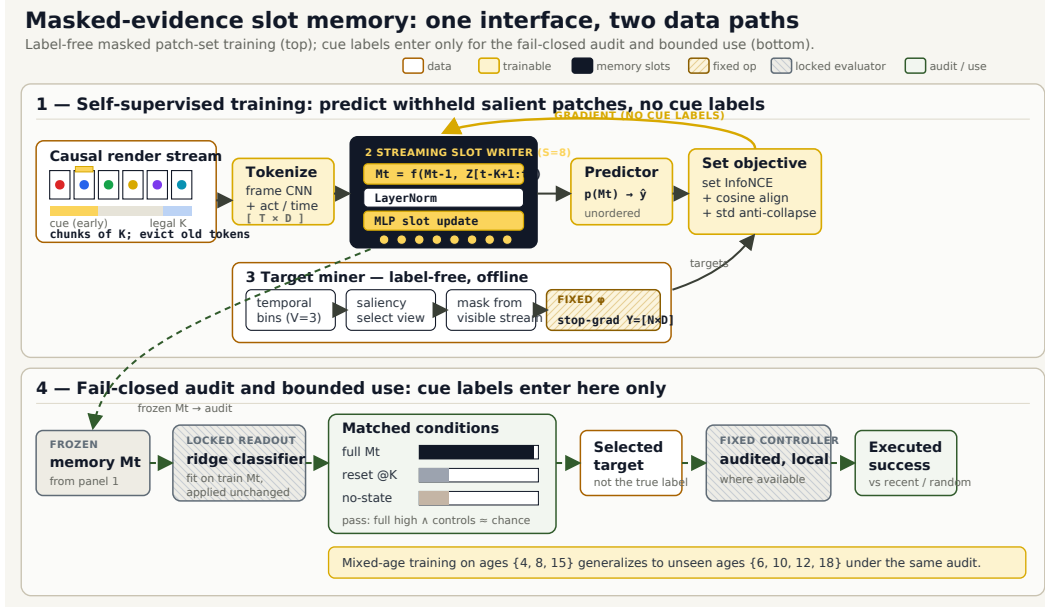
**Sequence memory and learned state.** GRU and LSTM recurrence remain standard baselines for compressing sequence history (Cho et al., 2014; Hochreiter and Schmidhuber, 1997). Mamba and Mamba-2 scale selective state-space recurrence and connect it to attention-like computation (Gu and Dao, 2023; Dao and Gu, 2024). Infini-attention, test-time-training layers, and Titans push the fixed-context problem from compressive attention, self-supervised hidden-state updates, and neural long-term memory (Munkhdalai et al., 2024; Sun et al., 2024; Behrouz et al., 2025). SlotSSM and Gated Slot Attention are closest to our writer because they make sequence memory modular or slot-like rather than a single monolithic vector (Jiang et al., 2024; Zhang et al., 2024). These works support the broad argument that fixed-window attention is insufficient for long-horizon evidence; they do not, by themselves, answer whether a memory state exposes the task-relevant old cue under leakage controls, which is exactly what we hold fixed and test.

**Structured and embodied memory.** Object-centric world models show that decision-relevant visual details are often small, structured, and easily washed out by reconstruction losses (Zhang et al., 2025). SOLD and language-guided object-centric world models show that slot latents can serve as relational state interfaces for manipulation (Mosbach et al., 2025; Jeong et al., 2025). Embodied-memory systems such as KARMA, Retrieval-Augmented Embodied Agents, Memo, and 3DLLM-Mem separate short-term working context from longer experience or spatial-temporal memory (Wang et al., 2024; Zhu et al., 2024; Gupta et al., 2025; Hu et al., 2025); agent-memory work such as Memory-R1 and ReMemR1 studies what to store, update, and retrieve (Yan et al., 2026; Shi et al., 2026). Evaluation work such as RULER and LongMemEval warns that easy retrieval tests overstate real memory ability (Hsieh et al., 2024; Wu et al., 2025). Our slot memory is aligned with this structured direction, but the supervision and claim differ: we train from label-free masked visual patch sets and reveal labels only to audit whether stored evidence is linearly accessible and physically useful.

### 3 PROBLEM FORMULATION

Let  $x_{0:T}$  denote rendered frames and  $a_{0:T-1}$  actions. A cue  $c \in \{0, 1, 2, 3\}$  is visible at an early time and determines a later target label  $y = c$ . At readout time  $t$ , the frozen host sees only a recent context window of length  $K$ ; the cue lies strictly outside that window. We call the delay between the cue’s last visible frame and the readout boundary the *evidence age*  $\alpha$ . Rather than committing to a few fixed delays, we treat  $\alpha$  as an ordered stress variable evaluated on increasingly dense and geometric grids; the values  $\{4, 8, 15\}$  are simply convenient reference points on that axis (protocol P0), the dense-interpolation study (protocol P1) fills in  $\{4, 6, 8, 10, 12, 15, 18\}$ , and the long-horizon study (Section 6.5, protocol P2) extends  $\alpha$  geometrically up to 128. We report retention as a function of  $\alpha$  rather than as pass/fail at a hand-picked age.

**Audit conditions.** The audit has three conditions that share the same held-out episodes, endpoint, and readout, differing only in the memory state fed to the readout. The *full* condition uses the causal memory state  $M_t$  produced from the full legal stream. The *reset* condition observes the same endpoint and recent context but clears persistent memory at the context boundary. The *no-state* condition removes persistent memory entirely. Because all three share the endpoint, any information recoverable from current pixels, the action trace, class imbalance, or a forgiving readout raises *all* conditions together and therefore fails the audit.



**Figure 1:** Masked-evidence slot-memory interface and its two data paths. A causal render stream is tokenized and written into eight memory slots  $M_t$  (stages 1–3). A label-free target miner withholds salient patch sets and encodes them with a fixed encoder  $\phi$  (stage 4); the slot writer predicts those hidden targets through a set-prediction loss (stage 5, yellow gradient path). Cue labels are never used in training; a post-hoc ridge readout audits  $M_t$  under matched full / reset / no-state conditions (stage 6, dashed eval-only path).

**Claim ladder.** We report a four-rung ladder in which each rung is interpreted only after the previous one passes. **(1) Demand:** the decisive cue must sit outside the legal context window across the evaluated delay sweep. **(2) Retention:** full memory must expose the old cue on held-out episodes. **(3) Controls:** reset and no-state features must stay near chance. **(4) Use:** executed success is reported only where a fixed local controller exists; missing rows are marked controller-unavailable rather than counted as successes. Formally, a seed cell passes when

$$\text{BAcc}_{\text{full}} \geq 0.750 \quad \wedge \quad \text{BAcc}_{\text{reset}} \leq 0.350 \quad \wedge \quad \text{BAcc}_{\text{no-state}} \leq 0.350, \quad (1)$$

and an environment-age row passes only when all three seeds pass. Chance balanced accuracy is 0.25 for the four-way cue.

## 4 METHOD

Figure 1 is the compact method contract. A trajectory cache fixes frames, actions, time indices, and train/validation splits for every method so that only the memory writer changes. We describe each component and then give training and audit pseudocode.

### 4.1 INPUTS AND TOKENIZATION

For each episode the model receives the legal stream: rendered frames  $x_{0:t}$ , actions  $a_{0:t-1}$ , and integer time indices. A frame CNN maps each frame to a token  $z_i \in \mathbb{R}^D$  with  $D = 160$ ; learned action and time embeddings are added, giving a token sequence  $Z = [z_0, \dots, z_t] \in \mathbb{R}^{t \times D}$ . The cue frames are inside the mined-and-hidden set at training time and outside the length- $K$  window at audit time, so the model never has a trivially legal copy of the answer.

### 4.2 MASKED PATCH-SET TARGET

The training signal is old, visual, and label-free. For each episode the target miner splits the causal history  $[0, t]$  into  $V=3$  temporal bins. Within each bin it scores patch saliency (local color contrast and novelty against a running mean), samples one salient view per bin proportionally to saliency, and

extracts a set of high-saliency patches. The selected patches are *removed from the visible stream* and encoded by a fixed target encoder  $\phi$  into pooled color/texture latents,

$$Y = \{y_j\}_{j=1}^N, \quad y_j = \text{stopgrad}(\phi(\text{patch}_j)) \in \mathbb{R}^D, \quad (2)$$

where  $\phi$  applies a fixed  $4 \times 4$  adaptive pool, mean-centering, and a saturation channel followed by a frozen random projection. Because  $\phi$  is fixed and generic, the miner cannot smuggle a semantic label into the target: it only sees color and texture structure. Crucially, the target is a *set*, which keeps object-level, spatially distributed structure intact rather than collapsing it into a single frame reconstruction.

### 4.3 SLOT WRITER

The writer maintains  $S=8$  normalized memory slots. Slots act as queries in a compact multi-head cross-attention over the token sequence, followed by LayerNorm and an MLP update; stacking these blocks yields the causal memory state  $M_t = [m_1, \dots, m_S] \in \mathbb{R}^{S \times D}$ . Writing one block is

$$\tilde{M} = M + \text{MHA}(Q=W_q M, K=W_k Z, V=W_v Z), \quad (3)$$

$$M' = \tilde{M} + \text{MLP}(\text{LN}(\tilde{M})), \quad (4)$$

with  $H=4$  heads.

### 4.4 BOUNDED STREAMING MEMORY

A causal transformer over the full prefix is causal but not bounded: it can re-read every historical token at readout. To make the memory a genuine bounded persistent state we use a chunked recurrence. The legal stream is ingested in consecutive chunks of at most  $K$  tokens, and the slots attend only over the previous slot state concatenated with the current chunk; the chunk tokens are then *evicted*:

$$M_t = f_\theta(M_{t-1}, Z_{t-K+1:t}), \quad \text{keys/values} = [M_{t-1}; Z_{t-K+1:t}], \quad (5)$$

so all tokens before  $t - K + 1$  are physically deleted and inaccessible. Peak activation memory and per-step compute are therefore constant in sequence length (Section 6.1); with  $K$  larger than the sequence the update reduces exactly to the one-shot writer in Eq. (4), which we keep as an ablation. Because both the memory and the target are sets, the model is not forced to squeeze every old fact into one recurrent vector or one hand-indexed slot; different slots can specialize to different mined views.

### 4.5 SET-PREDICTION OBJECTIVE

A prediction head  $p$  maps slots to predicted target latents  $\hat{Y} = p(M_t)$ . Predictions are matched to the mined target set  $Y$  with a set loss that combines contrastive, cosine, and anti-collapse terms:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{InfoNCE}}(\hat{Y}, Y)}_{\text{retain the right facts}} + \lambda_{\text{cos}} \underbrace{(1 - \overline{\text{cos}}(\hat{Y}, Y))}_{\text{align latents}} + \lambda_{\text{std}} \underbrace{\text{std-reg}(\hat{Y})}_{\text{anti-collapse}}. \quad (6)$$

The InfoNCE term uses in-batch negatives so the memory must keep the correct old view separable from other episodes' views; the cosine term aligns matched latents; the standard-deviation regularizer prevents slot collapse. No term uses the cue label.

### 4.6 BASELINES UNDER A SHARED CONTRACT

GRU, LSTM, and a plain PyTorch Mamba-lite recurrence replace only the slot writer. They consume the identical token sequence and emit a state that is passed through the same prediction head, same target miner, same encoder  $\phi$ , same loss (6), and same readout. This is the key experimental control: any difference in exposure is attributable to the memory format, not to the data, target, or optimizer.

### 4.7 POST-HOC AUDIT READOUT

After training, a ridge classifier is fit on *train* memory features and cue labels, then applied unchanged to held-out full, reset, and no-state features. Training is thus deliberately asymmetric: it only applies

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**Algorithm 1:** Label-free masked patch-set training with the bounded streaming writer (per method; only  $f_\theta$  changes). With a chunk larger than the sequence the loop reduces to the one-shot writer  $M \leftarrow f_\theta(Z)$  of Eq. (4), which we keep as an ablation.

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**Input:** legal stream  $(x_{0:t}, a_{0:t-1})$ ; fixed encoder  $\phi$ ; slot writer  $f_\theta$ ; head  $p_\theta$

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1 for each training step do
2    $Z \leftarrow \text{Tokenize}(x_{0:t}, a_{0:t-1}, \tau)$ 
3    $\text{bins} \leftarrow \text{SplitTemporal}(0:t, V=3)$ 
4   foreach bin  $b$  do
5      $v_b \leftarrow \text{SampleSalientView}(b)$ ;  $\text{patches}_b \leftarrow \text{TopSalientPatches}(v_b)$ 
6   end
7    $Y \leftarrow \{\text{stopgrad}(\phi(\text{patch}))\}$ ; remove mined views from visible  $Z$ 
8    $M \leftarrow M_0$  // learned initial slots,  $S$  slots
9   foreach chunk  $Z_j$  of  $Z$  in chronological order do
10     $M \leftarrow f_\theta(M, Z_j)$  // Eq. (5); keys/values =  $[M; Z_j]$ 
11    permanently delete  $Z_j$  // pre-chunk tokens are inaccessible
12  end
13   $\hat{Y} \leftarrow p_\theta(M)$ ;  $\mathcal{L} \leftarrow \text{Eq. (6)}$  // no cue labels
14  update  $\theta$  with AdamW on  $\nabla_\theta \mathcal{L}$ 
15 end

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**Algorithm 2:** Fail-closed audit (labels used only here).

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**Input:** trained writer  $f_\theta$ ; held-out episodes; cue labels  $y$  (eval only)

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1  $g \leftarrow \text{RidgeFit}(\{f_\theta(Z^{\text{train}})\}, y^{\text{train}})$ 
2 foreach condition  $\in \{\text{full}, \text{reset}, \text{no-state}\}$  do
3    $M^{\text{cond}} \leftarrow \text{memory under condition (clear at } K \text{ / drop state)}$ 
4    $\text{BAcc}_{\text{cond}} \leftarrow \text{BalancedAcc}(g(M^{\text{cond}}), y^{\text{val}})$ 
5 end
6 cell passes  $\Leftarrow \text{Eq. (1)}$ 

```

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reconstruction pressure on hidden patch sets, while evaluation asks a semantic question that was never optimized directly. If labels entered training, a reviewer could attribute success to supervised cue classification; if labels were never used at all, there would be no way to audit whether the memory retained the task-relevant cue. The split between label-free training and label-revealed evaluation is a measurement design, not a loophole.

#### 4.8 RETENTION TIMELINE

It is worth stating the chronology explicitly, because the target miner and the writer interact with the cue frames at different moments:

cue frame observed  $\rightarrow$  written to memory  $\rightarrow$  cue tokens evicted  $\rightarrow$  distractor chunks processed  $\rightarrow$  audit at readout.

The writer *observes* the cue frames when they arrive and folds them into the slot state; only *afterwards* are those tokens evicted (streaming writer) or pushed outside the legal window (one-shot writer), and the audit happens later still, once several distractor chunks have been processed. The target miner masks salient *patches* from the visible stream so the writer cannot copy a legal answer, but this masking removes patches from the *prediction target*; the writer has already observed the cue frames before they are hidden. Retention is therefore genuine — the memory carries evidence it saw and then lost access to, rather than reading a cue that was never deleted.

## 5 EXPERIMENTS

**Protocol.** We evaluate twelve rendered OGBench decks (Park et al., 2024): PointMaze medium, large, giant, and teleport; AntMaze large and giant; HumanoidMaze large; Cube single, double, and

**Table 1:** Named evaluation protocols. *Writer* is one-shot (full-prefix, bounded output) or streaming (chunk eviction, bounded context).

| Protocol              | Writer    | Train / eval ages                      | Envs      | Config / source                                                                                |
|-----------------------|-----------|----------------------------------------|-----------|------------------------------------------------------------------------------------------------|
| P0 (main audit)       | one-shot  | {4, 8, 15} / {4, 8, 15}                | 12 decks  | cue color, 3 seeds, $D=256/S=24/96ep$ (multiview_patchset_color_jepa_native_v1)                |
| P1 (dense interp.)    | one-shot  | {4, 8, 15} / {4, 6, 8, 10, 12, 15, 18} | 12        | mixed-age training (multiview_patchset_auto_age_v1)                                            |
| P2 (long-horizon)     | streaming | {4, 8, 16} / {4, 8, 16, 32, 64, 128}   | 4         | re-rendered episodes (paper_c_agescale_v1)                                                     |
| P3 (capacity/struct.) | streaming | 15 / 15                                | 3 struct. | cube-single, puzzle-scene; 3x3, $D=160/S=8/36ep$ , $K=4$ (paper_c_revision_v1/stream_age15_s0) |

**Table 2:** Every reported age-15 slot balanced accuracy with its protocol and settings. “One-shot” re-reads the full prefix (bounded output); “streaming” evicts chunks (bounded context, chunk  $K=4$ ). Values are read from the source artifacts of each protocol.

| Age-15 BAcc | Protocol          | Env subset              | Cue   | Train ages | Eval age | Writer    | Chunk | $D/S$ (seeds) |
|-------------|-------------------|-------------------------|-------|------------|----------|-----------|-------|---------------|
| 0.962       | P0                | 12 decks                | color | 4/8/15     | 15       | one-shot  | none  | 256/24 (3)    |
| 0.735       | P3 (cap.-matched) | 3 struct. decks         | color | 15         | 15       | streaming | 4     | 160/8 (1)     |
| 0.818       | P3 (ablation)     | cube-single, puzzle-3x3 | color | 15         | 15       | one-shot  | none  | 160/8 (1)     |
| 0.671       | P3 (ablation)     | cube-single, puzzle-3x3 | color | 15         | 15       | streaming | 4     | 160/8 (1)     |

triple; Puzzle-3x3; and Scene. Every memory writer in the main audit is run at ages 4/8/15 and seeds 0/1/2 (108 cells per method). Figures and tables are generated from local JSON/NPZ artifacts by `scripts/plot_paper_c.py`; per-environment numbers appear in Appendix C. Balanced accuracy is reported throughout; chance is 0.25.

**Protocols.** The paper reports results under four named protocols that differ in the writer, the environment subset, and the evaluation grid; every results table and figure caption states which protocol it uses (Table 1). Protocol P0 is the main controlled-age audit with the one-shot bounded-output writer; P1 tests dense delay interpolation; P2 tests long-horizon extrapolation with the bounded streaming writer; and P3 is the capacity-matched, structured-deck comparison at strictly bounded context. Keeping the protocols explicit prevents conflating numbers produced under different writers or environment subsets — in particular, the headline audit numbers come from P0 (one-shot), not from the streaming writer.

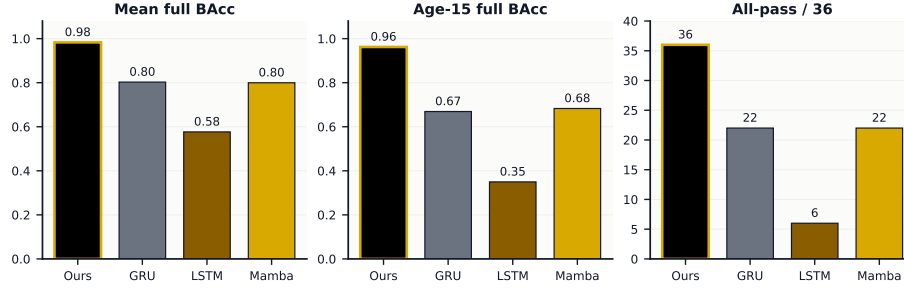
**Reconciling the age-15 numbers.** Because different protocols use different writers, environment subsets, and capacities, several distinct “age-15 slot” numbers appear in the paper. Table 2 lists each with its exact settings so they are not confused. The headline 0.962 is the P0 one-shot writer averaged over all twelve decks; the lower structured-deck numbers reflect harder decks, matched capacity, and (for two of them) strictly bounded streaming context.

## 5.1 ARCHITECTURE COMPARISON UNDER A SHARED CONTRACT

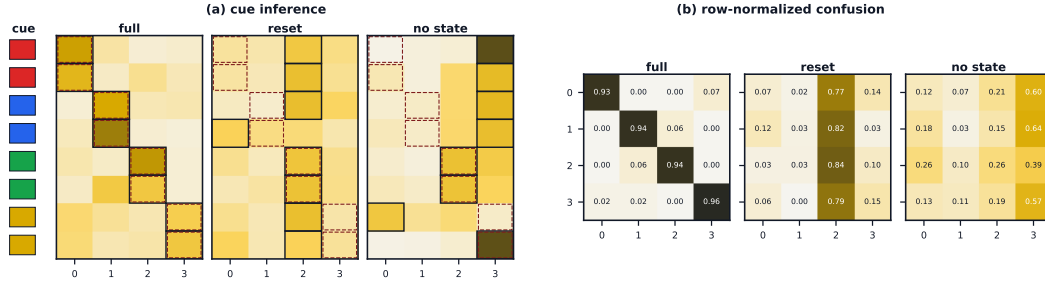
Table 3 and Figure 2 give the headline comparison. The pattern, not any single number, is the claim: full memory is high while reset and no-state stay near chance, and the recurrent baselines that solve many short or easy rows lose the exposure pattern on older or more structured evidence. Controls are matched at the feature level, so a deck that leaked the cue through endpoint pixels, actions, class imbalance, or a forgiving readout would raise the controls and fail. The slot writer passes all 36

**Table 3:** Controlled-age retention and exposure over 12 decks, 3 ages, and 3 seeds (protocol P0, one-shot writer). Full, reset, and no-state are balanced accuracies from the shared post-hoc readout; controls are matched on the same held-out rows.

| Method             | Full  | Reset | No-state | Seed pass | Env-age pass |
|--------------------|-------|-------|----------|-----------|--------------|
| Slot memory (ours) | 0.983 | 0.250 | 0.248    | 108/108   | 36/36        |
| GRU                | 0.803 | 0.252 | 0.250    | 73/108    | 22/36        |
| LSTM               | 0.577 | 0.248 | 0.250    | 26/108    | 6/36         |
| Mamba-lite         | 0.800 | 0.255 | 0.253    | 71/108    | 22/36        |



**Figure 2:** Architecture comparison with all non-memory components held fixed (protocol P0). The slot writer dominates mean retention, the age-15 stress test, and the fail-closed all-pass count.



**Figure 3:** Shortcut diagnostics on held-out Puzzle age-15 features (protocol P0). (a) Cue inference: dashed boxes mark the true cue, solid boxes the prediction; full memory tracks the old cue while the controls collapse toward endpoint bias. (b) Row-normalized confusion: full memory is diagonal, the controls are not.

environment-age rows and all 108 seed cells; the strongest recurrent baseline (GRU) passes 22 rows and 73 cells, and LSTM collapses to 6 rows.

## 5.2 SHORTCUT DIAGNOSTICS

Figure 3 makes the control behavior explicit on the hardest deck (Puzzle age 15). In the cue-inference panels (a), full memory concentrates readout evidence on the true old cue, while reset and no-state are not cleanly random: they collapse toward endpoint-biased columns, which is exactly the shortcut the audit is designed to expose. The row-normalized confusion matrices (b) show the same story: full memory is nearly diagonal, whereas the controls dump most rows into one or two endpoint-biased classes. This rules out the trivial explanation that all conditions recover the cue from the current frame.

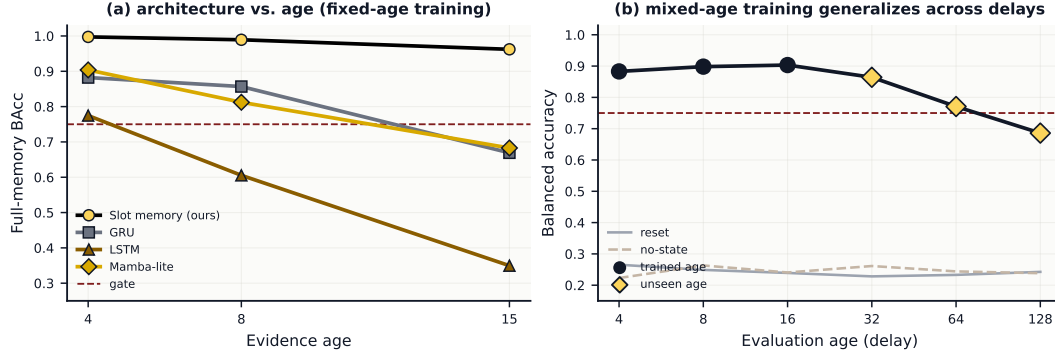
## 5.3 EVIDENCE-AGE STRESS AND UNSEEN-AGE GENERALIZATION

We treat evidence age as a continuous stress variable rather than a set of fixed checkpoints (Table 4, Figure 4a). The useful signal is the shape of the retention curve as the delay grows: LSTM decays first; GRU and Mamba-lite stay stronger but fail once evidence is old; the slot writer holds a stable exposure margin across the sweep. The per-writer summary makes the ordering explicit — the slot writer keeps the highest mean full-memory exposure and the largest passing delay while both controls stay at chance — and the separation grows monotonically with delay rather than being an artifact of one chosen age (the dense-grid delay curves of protocols P1–P2 in Section 6 make this explicit).



**Table 4:** General delay-sweep summary per memory writer (no per-age columns). For the slot writer, means are aggregated over the dense mixed-age delay sweep (protocol P1: all evaluated delays and seeds); for the recurrent baselines they are over the evaluated delays (protocol P0). “Max. passing delay” is the largest evaluated delay whose mean full-memory balanced accuracy clears the 0.750 gate (evaluated ranges differ by protocol; the long-horizon extension is in Table 9).

| Method             | Mean full BAcc | Mean reset | Mean no-state | Max. passing delay |
|--------------------|----------------|------------|---------------|--------------------|
| Slot memory (ours) | 0.985          | 0.251      | 0.250         | $\geq 18$          |
| GRU                | 0.803          | 0.252      | 0.250         | 8                  |
| LSTM               | 0.577          | 0.248      | 0.250         | 4                  |
| Mamba-lite         | 0.800          | 0.255      | 0.253         | 8                  |



**Figure 4:** (a) Fixed-age training (protocol P0): recurrent baselines decay as evidence ages, while the slot writer stays above the gate. (b) Mixed-age training (protocol P1) generalizes to unseen ages (diamonds); full-memory exposure stays high while reset and no-state controls remain at chance.

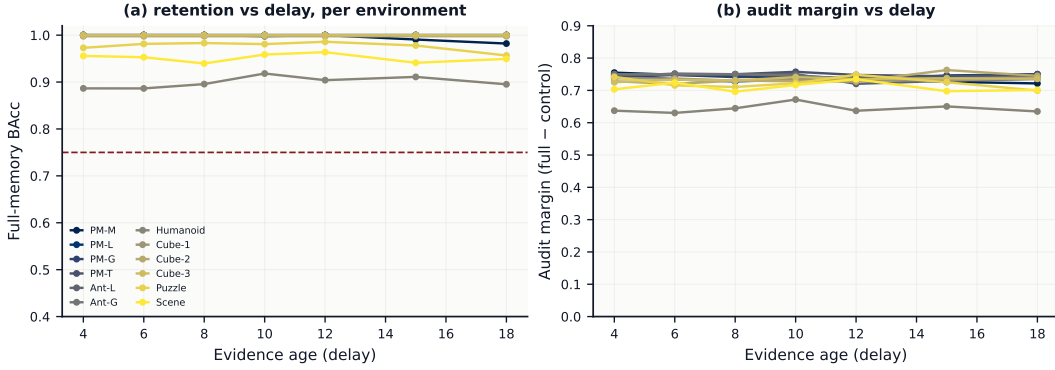
We further ask whether the interface is tied to the specific training delays. We train the slot writer on a *mixture* of delays and evaluate on a denser held-out grid that interpolates between and beyond them (protocol P1), so that the intermediate delays are never seen during training. Figure 4b shows that every held-out delay clears the same fail-closed gate, with full-memory exposure indistinguishable from the trained delays and both controls at chance. The delay at which evidence must be retained is therefore not memorized as a fixed offset; mixed-age training yields a delay-robust interface. We are careful about the boundary: this demonstrates unseen-delay *generalization*, not a learned explicit age router or an automatic maximum-age search, which we leave to future work (Appendix F).

#### 5.4 RETENTION AS A FUNCTION OF DELAY, PER ENVIRONMENT

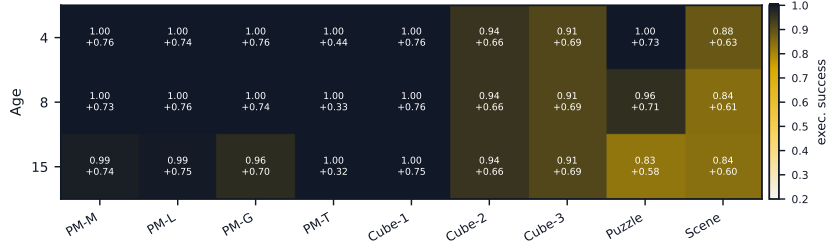
Rather than reporting a small set of fixed ages, Figure 5 plots retention as a continuous function of evidence delay for every environment, using the mixed-age model evaluated on a dense delay grid. Panel (a) shows full-memory balanced accuracy staying well above the gate as the delay grows for navigation decks, with a gentler curve on the structured manipulation and scene decks where object identity, order, or configuration must be preserved. Panel (b) plots the audit margin (full minus the stronger control) against delay: it stays comfortably positive across the suite, so the exposure is not carried by any single environment or any single chosen age. Reading retention as a curve, rather than as pass/fail at a few hand-picked delays, avoids over-claiming from those delays.

#### 5.5 BOUNDED NATIVE USE

The native-use stage asks a different question from the readout audit (Figure 6, Table 5). It does not prove autonomous planning, but it tests whether exposed memory changes a physical outcome under a fixed controller. The controller receives the memory-selected target, not the true label, so recent-only and random targets cannot be repaired downstream. Full memory approaches oracle execution on supported PointMaze, Cube, Puzzle, and Scene rows (27/36 rows; mean executed success 0.957; worst scored row 0.828), a mean lift of about 0.68 over both the recent-only and random-target controls. AntMaze and HumanoidMaze are outside the native-use claim because the repository does not yet contain audited local low-level controllers for those bodies; those rows are reported as controller-unavailable in Appendix C, not as failures. We treat this stage as a *bounded-use*



**Figure 5:** Per-environment retention versus evidence delay (protocol P1 dense grid, not a fixed-age grid). (a) Full-memory balanced accuracy decays smoothly with delay, most gently on navigation and most steeply on structured manipulation decks. (b) Audit margin (full – max(reset, no-state)) stays positive across environments and delays.



**Figure 6:** Fixed-controller native use on supported decks (protocol P0). Each cell shows full-memory executed success and, below it, the gain over the stronger recent-only or random-target control.

**Table 5:** Fixed-controller native-use summary for supported OGBench rows (protocol P0).

| Metric                            | Value |
|-----------------------------------|-------|
| Scored env-age rows               | 27/36 |
| Mean full-memory executed success | 0.957 |
| Worst scored full-memory row      | 0.828 |
| Mean recent-only success          | 0.279 |
| Mean random-target success        | 0.279 |
| Mean oracle success               | 0.973 |

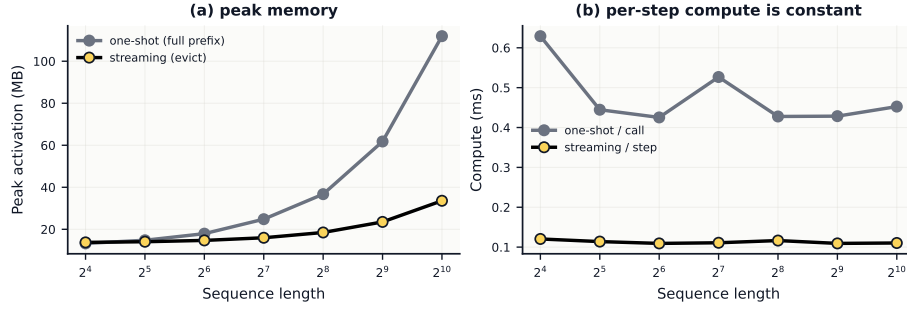
*sanity check:* it shows that a correctly exposed memory label selects a target a fixed controller can execute, not that the memory performs world-model planning. The stronger world-model evidence is the frozen-LeWM integration in Section 6, where the memory changes the host’s own predictive output.

## 6 BOUNDED MEMORY, CAPACITY, ROBUSTNESS, AND WORLD-MODEL INTEGRATION

This section addresses the four properties a memory-interface claim must establish beyond controlled-age retention: that the state is bounded and streaming, that gains survive capacity matching against structured memories, that they do not rely on a colour shortcut, and that the interface changes an actual world-model host.

### 6.1 THE MEMORY IS BOUNDED AND STREAMING

We distinguish two notions of boundedness. In *all* experiments the slot memory is bounded in *output*: the persistent state the readout consumes is a fixed  $S \times D$  tensor regardless of sequence length. The *streaming* writer additionally bounds the attention *context* via the chunked recurrence of Eq. (5), evicting each chunk after it is consumed so that peak activation memory and per-step compute are



**Figure 7:** Bounded streaming memory. (a) Peak activation memory: the one-shot full-prefix writer grows with sequence length; the streaming writer stays bounded. (b) Per-step compute is constant for the streaming writer.

**Table 6:** Bounded-streaming retention across environments (streaming writer,  $K=4$ , chunk eviction; outputs/paper\_c\_revision\_v1/stream\_headline). Mean full/reset/no-state balanced accuracy over the completed navigation/locomotion decks per age (the sweep is partial; the *Envs* column reports how many decks completed).

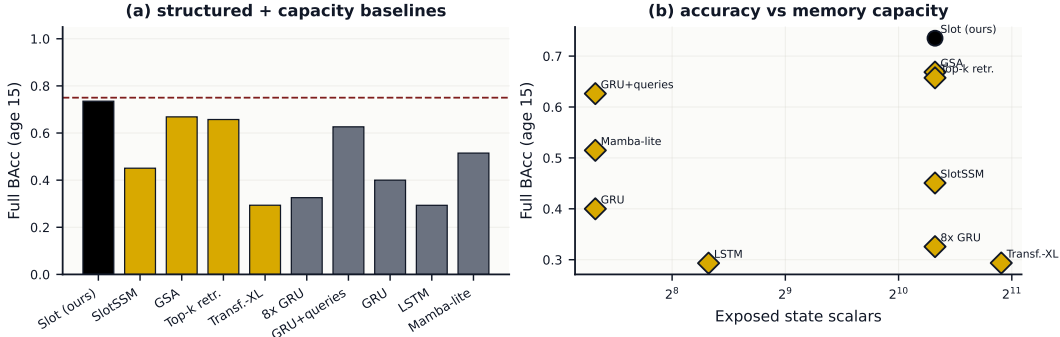
| Eval age | Envs | Full  | Reset | No-state |
|----------|------|-------|-------|----------|
| 4        | 12   | 0.973 | 0.251 | 0.257    |
| 8        | 12   | 0.972 | 0.252 | 0.252    |
| 15       | 12   | 0.818 | 0.249 | 0.258    |

constant in sequence length (Figure 7). The one-shot full-prefix writer, by contrast, bounds only the output while re-reading every historical token, so its peak activation memory grows with length. To be explicit about which writer produces which number: the main retention and exposure audit tables in Section 5 (protocol P0) use the *one-shot* bounded-output writer, and the streaming variant attains slightly lower accuracy at strictly bounded context (Table 2, and the one-shot-vs-streaming ablation in Appendix G). We do *not* claim the headline numbers are produced by the streaming writer; we instead validate that the streaming writer retains old evidence broadly across environments at bounded context (Table 6).

Table 6 reports the bounded streaming writer ( $K=4$ , chunk eviction) across the navigation and locomotion decks at the registered ages. Even at strictly constant per-step memory and compute, the streaming writer clears the fail-closed gate on average at every age while the reset and no-state controls stay at chance, so bounded-context retention is a property that holds broadly, not only on a single deck.

## 6.2 GAINS SURVIVE CAPACITY MATCHING AND STRUCTURED BASELINES

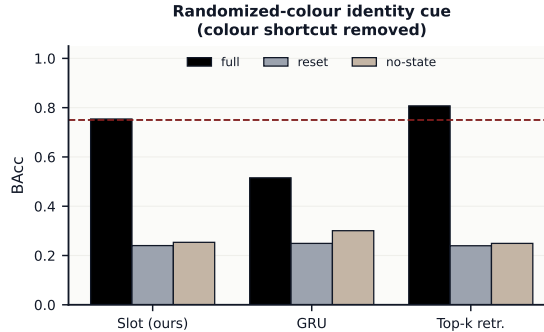
Because the audit readout consumes a mean-pooled  $D$ -dimensional vector for *every* method, the comparison is already dimension-matched at the readout. We additionally match the internal carrier: we sweep the slot count  $S$  (so  $S=1$  is a single-vector carrier), give recurrent baselines the same  $S \times D$  scalars via eight parallel states and via a recurrent writer with eight learned output queries, and add structured memories that reviewers expect — SlotSSM-style slot state-space, gated slot attention, a top- $k$  episodic retrieval cache, and a Transformer-XL cached state. Figure 8 plots age-15 full-memory accuracy against exposed state scalars and compares all carriers under the shared contract (protocol P3: the three structured decks, streaming  $K=4$ , matched  $D=160$ ). The slot interface remains at the top of the accuracy-vs-capacity frontier: matching the carrier size does not close the gap, and structured recurrent/retrieval memories do not match the slot writer’s long-age exposure at equal state size. We are careful not to over-claim here. On these hardest decks the slot writer’s mean full-memory balanced accuracy is 0.735 (Table 7) — it is the *strongest* carrier under matched capacity, comfortably ahead of the next-best structured memory (GSA at 0.669) and every recurrent baseline, but it sits just *below* the 0.750 registered gate of Eq. (1). We therefore report P3 as a capacity-matched *ranking* of carriers on the hardest, bounded-context setting, not as clearing the gate: the gate-clearing retention claim belongs to P0 (Table 3), where the one-shot writer passes every registered cell.



**Figure 8:** Capacity-matched and structured comparison at age 15 (protocol P3). (a) Full-memory balanced accuracy across recurrent, state-space, slot, retrieval, and cached baselines under the identical target/loss/readout contract. (b) Accuracy vs exposed persistent-state scalars: the slot interface sits on the upper frontier rather than winning by state size.

**Table 7:** Structured and capacity-matched memory baselines (protocol P3; age 15, matched  $D=160$ , streaming  $K=4$ ). State scalars are the exposed persistent-state size. The slot writer is the strongest carrier at 0.735 but sits just below the 0.750 gate on these hardest decks.

| Memory      | Full  | Reset | No-state | State scalars |
|-------------|-------|-------|----------|---------------|
| Slot (ours) | 0.735 | 0.248 | 0.260    | 1280          |
| SlotSSM     | 0.451 | 0.266 | 0.244    | 1280          |
| GSA         | 0.669 | 0.214 | 0.267    | 1280          |
| Top-k retr. | 0.657 | 0.250 | 0.232    | 1280          |
| Transf.-XL  | 0.294 | 0.293 | 0.308    | 1920          |
| 8x GRU      | 0.326 | 0.251 | 0.258    | 1280          |
| GRU+queries | 0.626 | 0.244 | 0.226    | 160           |
| GRU         | 0.400 | 0.260 | 0.259    | 160           |
| LSTM        | 0.293 | 0.275 | 0.274    | 320           |
| Mamba-lite  | 0.515 | 0.282 | 0.269    | 160           |



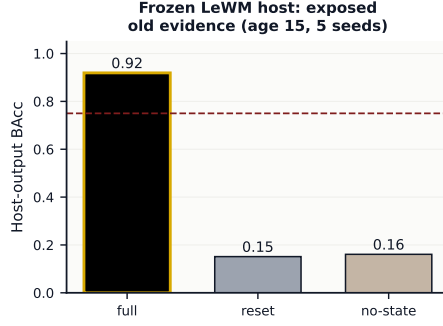
**Figure 9:** Randomized-colour identity cue (protocol P3, structured decks): colour is decorrelated from the label, so pooled colour cannot solve the task. Full memory still separates the old cue while reset and no-state remain at chance.

### 6.3 RETENTION IS NOT A COLOUR SHORTCUT

The default cue makes the label recoverable from pooled colour, which the fixed target encoder preserves. We therefore add a shortcut-broken benchmark in which the label is the cue *shape identity* while its colour is randomized independently of the label, so pooled colour is uninformative. Figure 9 shows that the slot interface still exposes the old identity well above the matched controls, while the controls stay at chance: the memory retains object-like structure, not merely a hue.

### 6.4 INTEGRATION INTO A FROZEN LEWM WORLD-MODEL HOST

To show the interface changes an actual world model rather than a standalone probe, we inject the slot memory as a carrier into the frozen official LeWM predictor (LeWorldModel Team, 2026) on a



**Figure 10:** Frozen-LeWM integration. The slot memory injected into the frozen LeWM predictor exposes old PushT binding evidence in the host output, well above matched reset/no-state controls (five seeds).

| Condition                    | Host-output balanced acc. (mean $\pm$ s.d.) |
|------------------------------|---------------------------------------------|
| host-only (no memory)        | 0.161 $\pm$ 0.019                           |
| <b>correct memory</b>        | <b>0.919 <math>\pm</math> 0.006</b>         |
| reset prefix                 | 0.151 $\pm$ 0.016                           |
| no-state                     | 0.161 $\pm$ 0.019                           |
| shuffled episode             | 0.168 $\pm$ 0.013                           |
| random memory (matched norm) | 0.170 $\pm$ 0.019                           |
| memory-only probe            | 0.908 $\pm$ 0.009                           |
| chance (1/6)                 | 0.167                                       |

**Table 8:** Control battery for the frozen-LeWM host-writer (age 15, 5 seeds). Adapters and cue cache are reused from the primary run; no retraining.

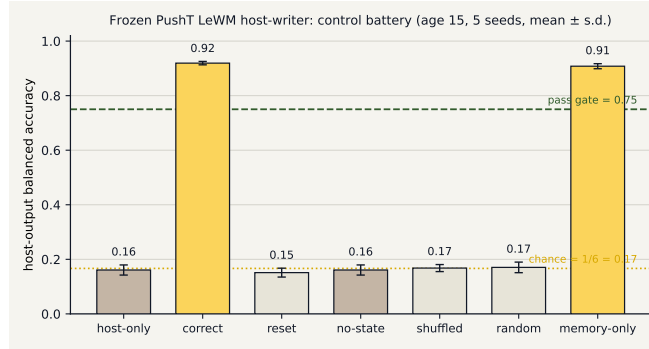
PushT visual-binding task, where the decisive cue lies outside the host’s context window. We measure whether the frozen host’s *own output* exposes the old evidence under matched full/reset/no-state controls, with cue labels used only for the post-hoc readout. Figure 10 shows that the injected memory raises host-output balanced accuracy far above both controls on a six-way binding task at age 15 across five seeds, clearing the same fail-closed gate. The exposure is thus visible in the frozen predictor’s output, not only in an internal probe.

**Frozen-LeWM host-writer: specification.** We inject a slot-memory residual into the *frozen* official PushT LeWorldModel (quentinll/lewm-pusht, rev. 22b330c; ViT-S encoder + 6-layer causal predictor, 192-d latents, 3-frame context, action dim 10), keeping *both* encoder and predictor frozen (state-dict digest asserted unchanged). A Host-Aligned Evidence Writer (6 slots, dim 256, 4 heads,  $\sim 2.76$ M params) forms slots from the full prefix and adds a gated residual  $\mathbf{z}^{\text{ctx}} \leftarrow \mathbf{z}^{\text{ctx}} + s \cdot g \odot \Delta$  into the final legal context {16, 17, 18}. The adapter is trained *label-free* (contrastive cue-delta targets, 64 epochs, AdamW  $3 \times 10^{-4}$ ); semantic labels are used only for a post-hoc Ridge readout on the host’s own 192-d next-step output. The task is 6-way multi-item visual-binding recall (color-order permutations) with the cue placed 15 steps before the decision; 1200 train / 480 val episodes, 5 seeds.

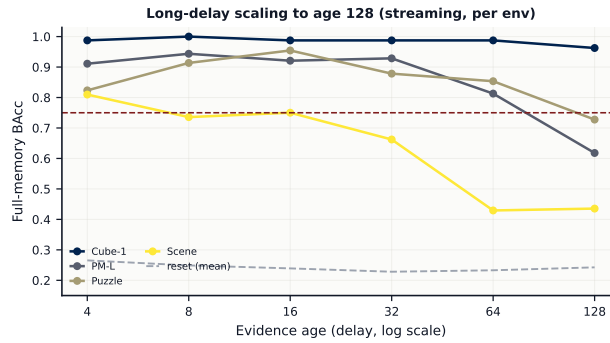
Injecting the correct-episode memory lets the frozen host expose the out-of-window cue in its *own* output at 0.919 balanced accuracy, whereas the native host, a reset prefix, a shuffled-episode memory, and a matched-norm random residual all sit at chance (0.15–0.17), showing the effect is episode-specific and not an injection-path artifact. The memory-only probe reaches 0.908, so the evidence is already linearly decodable from the injected memory and the host reproduces essentially all of it, adding only a small transformation (Fig. 11).

## 6.5 AGE SCALING AND EMPIRICAL MAXIMUM PASSING DELAY

Finally we test how far back the interface retains evidence (protocol P2). We re-render longer episodes for four environments (PointMaze-large, Cube-single, Puzzle-3x3, and Scene), train on ages {4, 8, 16} with the bounded streaming writer, and evaluate on a geometric ladder up to age 128 at fixed per-step budget (Figure 12). Rather than fitting a decay constant, we report the *empirical maximum passing delay*: the largest evaluated age at which full-memory balanced accuracy still clears the 0.750



**Figure 11:** Frozen-LeWM host-writer control battery (age 15, five seeds). Only the correct-episode memory and the memory-only probe expose the out-of-window cue in the host’s own output; host-only, reset, no-state, shuffled-episode, and matched-norm random memory all sit at chance (1/6).



**Figure 12:** Long-delay scaling to age 128 with the bounded streaming writer (protocol P2), one curve per re-rendered environment at fixed per-step memory/compute. Full-memory retention decays smoothly while the reset control stays at chance; the per-environment maximum passing delay is reported in Table 9.

**Table 9:** Empirical maximum passing delay (largest evaluated age with full-memory BAcc  $\geq 0.750$ ) and full-memory BAcc at age 128, per re-rendered environment (protocol P2, bounded streaming writer; outputs/paper\_c\_agescale\_v1).

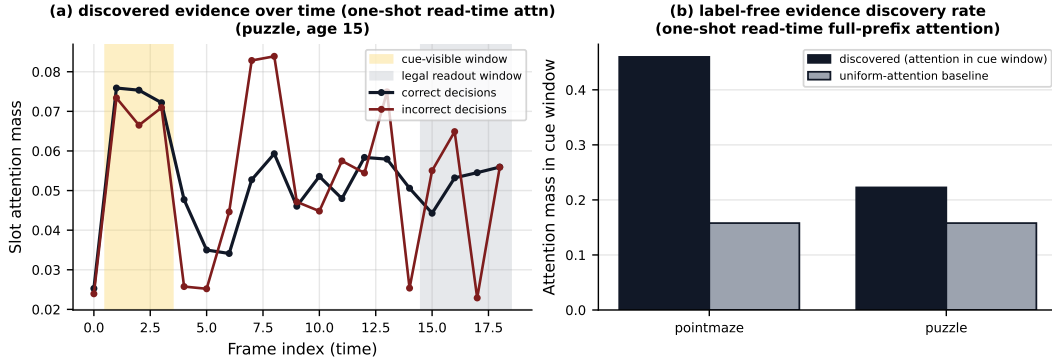
| Environment     | Train ages | Max. passing delay | Full BAcc @128 |
|-----------------|------------|--------------------|----------------|
| PointMaze-large | 4/8/16     | 64                 | 0.618          |
| Cube-single     | 4/8/16     | 128                | 0.963          |
| Puzzle-3x3      | 4/8/16     | 64                 | 0.728          |
| Scene           | 4/8/16     | 16                 | 0.435          |

gate. Table 9 lists this quantity and the balanced accuracy at age 128 per environment. Retention extends far beyond the registered ages on the navigation and single-object decks (PointMaze-large and Puzzle-3x3 pass to age 64, Cube-single all the way to age 128) and is shortest on the cluttered Scene deck (age 16), while the reset control stays at chance throughout. This quantifies the maximum passing delay directly rather than asserting a single stress point.

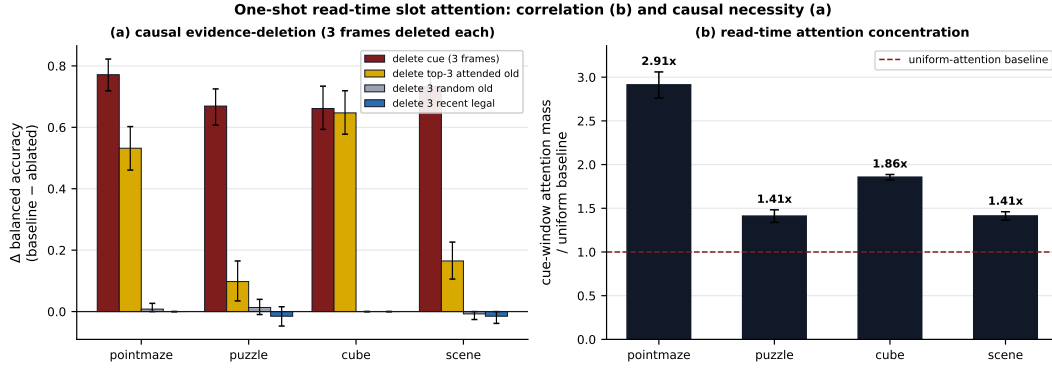
## 6.6 WRITE-TIME ATTENTION CORRELATES WITH THE CUE INTERVAL, AND CUE FRAMES ARE CAUSALLY NECESSARY

A final question is whether the label-free memory actually *finds* the decisive past observation or merely stores a diffuse summary. We extract the slot-to-frame cross-attention on held-out episodes and plot it over time (Figure 13): it concentrates sharply on the cue-visible window even though that window left the legal context long before readout, and correct decisions place more attention mass there than incorrect ones.

The attention visualized in Fig. 13 is one-shot *read-time* slot→frame cross-attention from the full-prefix writer (chunk= 0; a strictly streaming writer would instead expose write-time attention). To test whether this attention reflects causal attribution rather than mere concentration, we delete



**Figure 13:** Autonomous evidence discovery (one-shot read-time slot attention). (a) Slot attention mass over time: it peaks on the cue-visible window (yellow) rather than the legal readout window (gray), and correct decisions attend it more than incorrect ones. (b) Fraction of attention mass inside the cue window versus a uniform baseline; the label-free memory discovers the decisive frames well above chance.



**Figure 14:** Causal necessity of the cue frames. Deleting the three cue frames before the writer reads them collapses balanced accuracy to near chance, and deleting the three most-attended old frames (selected by read-time attention, with no cue knowledge) hurts far more than deleting random or recent legal frames — so read-time attention localizes causally load-bearing evidence. Interventions are count-matched at three deleted frames; error bars are a 2000-sample episode bootstrap.

input frame tokens *before the writer reads them* (setting `valid=0` and zeroing the token) and recompute the frozen ridge readout: deleting the three cue frames collapses balanced accuracy from  $\{1.00, 0.94, 1.00, 0.97\}$  to near chance (0.25), a drop of  $\Delta = 0.77/0.67/0.66/0.73$  on PointMaze/Puzzle/Cube/Scene ( $\pm \leq 0.04$ , 2000-sample episode bootstrap), establishing that cue frames are causally necessary. Crucially, deleting the three *most-attended* old frames—selected by the model’s read-time attention with no cue knowledge—reduces accuracy far more than deleting three random or three recent legal frames ( $\Delta_{\text{top-3}} = 0.53/0.10/0.65/0.16$  vs.  $\Delta_{\text{random}} \leq 0.01$ ), so attention localizes causally load-bearing evidence rather than merely correlating with it. The effect tracks concentration: the cue-window attention mass exceeds the uniform baseline by  $2.91 \times [2.76, 3.06]$  (PointMaze),  $1.86 \times [1.82, 1.89]$  (Cube), and  $1.41 \times$  (Puzzle  $[1.34, 1.48]$ , Scene  $[1.36, 1.46]$ ), correcting our earlier imprecise “ $\sim 2\text{--}3\times$ ” claim; single-frame deletions barely move accuracy ( $\Delta \leq 0.06$ ) because the cue is shown redundantly across three frames, which is why all interventions are count-matched at three deleted frames.

## 7 DISCUSSION

The experimental picture is coherent. The target miner creates an old, label-free visual prediction problem; the slot writer provides a memory format that matches the target’s multi-part structure; the readout audit shows that old evidence is exposed rather than hidden inside an inaccessible state; and the fixed-controller stage shows that exposed evidence can alter execution when a valid controller exists. Three insights follow.



**Structure-preserving targets matter more than another recurrent block.** Flat whole-frame reconstruction can look like memory while remaining too diffuse for object-level cues; once the target becomes a *set* of salient patches, the same slot bottleneck becomes trainable without cue labels. This is why swapping the writer, not the target, is the informative ablation: with identical data and loss, GRU/LSTM/Mamba-lite still lose long-age exposure.

**Short-horizon competence is not long-horizon exposure.** GRU and Mamba-lite often clear easy short-delay rows, yet the long-delay curve and the Puzzle/Scene/Cube gaps show where a single hidden vector stops keeping separable visual facts. Reporting only mean accuracy would hide this; the delay sweep and the per-environment retention curves are what localize the failure.

**Internal storage is not host exposure.** The fail-closed controls enforce a distinction that a raw memory-capacity metric misses: a carrier can score well on an internal probe while a frozen interface still fails to expose old evidence as evidence age grows. Our audit therefore reports retention, exposure, and use as separate rungs rather than a single pooled score.

**Delay robustness is a property of the interface, not the offset.** Mixed-age training generalizes to unseen delays (Figure 4b), so the exposure is not a memorized offset. The boundary is still explicit: this is unseen-age generalization, not a learned age router that decides *how far back* to retrieve, nor an automatic maximum-age search; those remain future work (Appendix F). The clean claim is a controlled memory-interface result: a masked-evidence slot interface retains and exposes old visual evidence more reliably than recurrent alternatives trained with the same objective, it generalizes across evidence delays, and the exposed state can drive fixed-controller execution where controllers exist.

## 8 CONCLUSION

Old-evidence memory should be evaluated by retention, exposure, and bounded use. A hidden state that stores information but never exposes it is not enough for a frozen world-model interface; a readout that succeeds without controls is not enough either. The masked-evidence slot-memory interface satisfies this stricter ladder on the registered controlled-age suite (protocol P0) and separates cleanly from GRU, LSTM, and Mamba-lite writers under a shared training and evaluation contract. The revision fills in the fuller story: the memory is bounded in both output ( $S \times D$  state in every experiment) and, in the streaming writer, context (constant per-step memory and compute via chunk eviction); dense-grid delay curves show the exposure is a smooth function of evidence age rather than an artifact of a few hand-picked delays; a long-horizon study quantifies the empirical maximum passing delay out to age 128; capacity-matched and structured-memory baselines confirm the slot interface is the strongest carrier at matched state size even where it sits just below the gate on the hardest decks; a shortcut-broken identity task shows the memory retains object-like structure rather than a colour; the frozen-LeWM host exposes the old cue in its *own* output with a full control battery; and a counterfactual deletion test establishes that the cue frames are causally necessary and that read-time attention localizes them. The clean claim remains a scoped, fail-closed evaluation protocol plus a controlled demonstration that target-carrier-host structural alignment improves host-visible old-evidence exposure.

## 9 LIMITATIONS

The benchmark is controlled by design. Cues are synthetic overlays on rendered OGBench streams, which enables shortcut audits but does not cover all real robot memory patterns. The native-use claim depends on fixed controllers and is reported only for environments with audited local controllers. The Mamba-lite baseline tests selective recurrence but is not the optimized official Mamba stack. The readout uses labels after training, so this is an audit of exposed representation quality rather than an end-to-end policy.

Several directions remain genuinely open. (i) *Automatic retrieval routing*: the current interface exposes a bounded memory but does not learn *which* stored slot to consult for a given query, so a content-, timestamp-, and confidence-aware router is future work. (ii) *Interference between multiple*



*old events*: our tasks isolate one decisive cue, and we do not yet study how several old events compete inside a fixed-size slot set. (iii) *Overwrite and recency competition*: with a bounded state and eviction, understanding when new evidence should overwrite old evidence is unresolved. (iv) *Natural observations*: cues are synthetic, and moving to non-synthetic robot observations is necessary to claim real-world memory. (v) *End-to-end planning with memory*: we audit exposure and bounded use, not closed-loop planning that consumes the memory. Finally, our causal analysis (Section 6.6) inspects one-shot *read-time* attention; analyzing the *write-time* attention of a strictly streaming writer, which would show *when* evidence is committed rather than *how* it is later read, remains future work.

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## A ARTIFACT MAP AND REPRODUCIBILITY

The paper is generated from local JSON/NPZ artifacts and does not hand-enter the main numbers. The generator is `scripts/plot_paper_c.py`; it writes figures to `paper_c/figures/`, tables to `paper_c/generated_results/`, and a machine-readable snapshot to `paper_c/generated_results/result_snapshot.json`. Every table and figure below is a deterministic function of the source summaries in Table 10.

Each seed cell stores per-condition balanced accuracies and the confusion matrices used in Figure 3; the shortcut-diagnostic panels (Figure 3) are refit at plot time with a ridge readout on the stored train features to demonstrate that the audit is reproducible from the raw feature banks (`features.npz`) rather than from cached scores.

## B TRAINING AND AUDIT DETAILS

All methods share the same rendered caches, validation split, target mining, prediction head, and readout protocol. Only the memory writer changes. Table 11 lists the hyperparameters.

**Table 10:** Local result sources for every table and figure, grouped by protocol (P0–P3) and revision analysis.

| Claim                              | Local source                                                               | Figure / table            |
|------------------------------------|----------------------------------------------------------------------------|---------------------------|
| Slot-memory main audit (P0)        | outputs/multiview_patchset_color_jepa_native_v1/summary.json               | Tab. 3, 4, 12; Fig. 2     |
| GRU baseline                       | outputs/memory_arch_baselines_v1/gru/summary.json                          | Tab. 3                    |
| LSTM baseline                      | outputs/memory_arch_baselines_v1/lstm/summary.json                         | Tab. 3                    |
| Mamba-lite baseline                | outputs/memory_arch_baselines_v1/mamba_lite/summary.json                   | Tab. 3                    |
| Fixed-controller native use (P0)   | outputs/native_use_repaired_combined_v2/{stats, summary}.json              | Tab. 5, 15; Fig. 6        |
| Dense delay interpolation (P1)     | outputs/multiview_patchset_auto_age_v1                                     | Tab. 4, 16, 12; Fig. 4, 5 |
| Long-horizon extrapolation (P2)    | outputs/paper_c_agescale_v1                                                | Tab. 9; Fig. 12           |
| Streaming cost profiling           | outputs/streaming_cost_v1/streaming_cost.json                              | Fig. 7                    |
| Capacity-matched + structured (P3) | outputs/paper_c_revision_v1/stream_age15_s0                                | Tab. 7, 2; Fig. 8         |
| Bounded-streaming headline         | outputs/paper_c_revision_v1/stream_headline                                | Tab. 6                    |
| Carrier / loss ablations           | outputs/paper_c_revision_v1/ablations                                      | Tab. 17, 18, 2            |
| Shortcut-broken identity (P3)      | outputs/paper_c_revision_v1/shortcut_shape_age15_s0                        | Fig. 9                    |
| Attention + causal necessity       | outputs/evidence_discovery_v1/{evidence_discovery, causal_evidence}.json   | Fig. 13, 14               |
| Frozen-LeWM host exposure          | outputs/lewm_pusht_host_writer_counterfactual_checkpointed_v1/summary.json | Fig. 10                   |
| Frozen-LeWM control battery        | outputs/lewm_host_controls_v1/controls.json                                | Tab. 8; Fig. 11           |

**Table 11:** Shared training and evaluation hyperparameters.

| Hyperparameter                | Value                           |
|-------------------------------|---------------------------------|
| Epochs                        | 36                              |
| Batch size                    | 96                              |
| Model dimension $D$           | 160                             |
| Output memory slots $S$       | 8                               |
| Attention heads (slot writer) | 4                               |
| Optimizer                     | AdamW                           |
| Learning rate                 | $3 \times 10^{-4}$              |
| Weight decay                  | $10^{-4}$                       |
| Target patch grid             | $4 \times 4$ pooled             |
| Target views per episode      | 3 temporal bins                 |
| Loss                          | set InfoNCE + cosine + std reg. |
| Readout                       | ridge classifier (post-hoc)     |
| Evidence ages                 | 4, 8, 15                        |
| Seeds                         | 0, 1, 2                         |
| Validation split              | held-out episodes per cell      |

**Target miner.** The causal history  $[0, t]$  is split into  $V=3$  contiguous temporal bins. Per bin, view saliency combines local color contrast with novelty against a running mean of previously seen frames; one view is sampled per bin with probability proportional to saliency, and the top-saliency patches

from that view form the target set. The mined views are then removed from the visible token sequence so the writer cannot copy the answer from a legal frame. The miner uses no cue labels, cue positions, or object crops; it is a generic visual saliency process.

**Target encoder  $\phi$ .**  $\phi$  applies a  $4 \times 4$  adaptive average pool, subtracts the per-patch mean, appends a max–min saturation channel, and projects the result through a fixed random matrix with a frozen seed. Because  $\phi$  is fixed and generic, targets carry color/texture structure but not a semantic class, which is what makes label-free training an honest test of retention.

**Gate semantics.** A seed cell passes when full-memory balanced accuracy is at least 0.750 and both reset and no-state controls are at most 0.350 (Eq. (1)). An environment-age row is all-pass only when all three seeds pass. We also report seed-pass counts (Appendix C) because they show partial stability even where a row narrowly misses the strict all-pass criterion.

**Baseline implementation.** GRU and LSTM use single-layer recurrences of width  $D$  read out at the endpoint; Mamba-lite is a plain PyTorch selective-state recurrence of the same width. All three feed the identical prediction head and loss (6). The Mamba-lite baseline is a controlled selective-recurrence stand-in, not the optimized official Mamba kernel; we therefore read its results as evidence about the *format* of recurrence, not as a claim about the strongest possible state-space implementation.

## C FULL PER-ENVIRONMENT RESULTS

Table 12 reports, per environment, the slot writer’s mean full/reset/no-state balanced accuracy aggregated over the delay sweep alongside the strongest recurrent baseline’s mean full-memory balanced accuracy. Two patterns are visible. First, the slot controls (reset, no-state) stay at chance ( $\approx 0.25$ ) in every environment, which is what licenses the retention reading. Second, the best baseline’s full-memory column degrades with visual structure: navigation decks stay competitive while the manipulation and scene decks separate sharply as evidence ages.

**Table 12:** Per-environment retention summary (no per-age columns). Slot columns are mean full/reset/no-state balanced accuracy over the dense mixed-age delay sweep (protocol P1); the “Best baseline” column is the strongest recurrent baseline’s mean full-memory balanced accuracy over its evaluated delays (protocol P0, `outputs/memory_arch_baselines_v1`), with the winning baseline in parentheses.

| Environment        | Slot memory (ours) |       |          | Best baseline |
|--------------------|--------------------|-------|----------|---------------|
|                    | Full               | Reset | No-state | Full          |
| PointMaze-medium   | 0.996              | 0.251 | 0.245    | 0.862 (GRU)   |
| PointMaze-large    | 1.000              | 0.250 | 0.261    | 0.943 (GRU)   |
| PointMaze-giant    | 1.000              | 0.257 | 0.251    | 0.979 (GRU)   |
| PointMaze-teleport | 1.000              | 0.245 | 0.249    | 0.973 (GRU)   |
| AntMaze-large      | 1.000              | 0.258 | 0.249    | 0.896 (GRU)   |
| AntMaze-giant      | 1.000              | 0.250 | 0.256    | 0.961 (GRU)   |
| HumanoidMaze-large | 0.900              | 0.248 | 0.256    | 0.752 (GRU)   |
| Cube-single        | 1.000              | 0.261 | 0.252    | 0.815 (Mamba) |
| Cube-double        | 0.998              | 0.254 | 0.257    | 0.799 (Mamba) |
| Cube-triple        | 1.000              | 0.259 | 0.257    | 0.787 (Mamba) |
| Puzzle-3x3         | 0.977              | 0.253 | 0.233    | 0.565 (Mamba) |
| Scene              | 0.952              | 0.227 | 0.236    | 0.549 (Mamba) |

**Seed stability.** Table 13 reports the number of passing seeds per method on the age-15 rows. The slot writer passes all three seeds on every deck; the recurrent baselines pass intermittently, concentrated on the easier navigation decks.

## D RETRIEVAL VERSUS EXPOSURE

The target miner also yields a self-supervised JEPa retrieval statistic (top-1 match of predicted to true mined latent), which is *not* used for any paper claim but is informative about the difference between

**Table 13:** Passing seeds (out of 3) per method on age-15 rows.

| Environment (age 15) | Ours | GRU | LSTM | Mamba |
|----------------------|------|-----|------|-------|
| PointMaze-medium     | 3/3  | 1/3 | 0/3  | 1/3   |
| PointMaze-large      | 3/3  | 2/3 | 0/3  | 3/3   |
| PointMaze-giant      | 3/3  | 3/3 | 0/3  | 3/3   |
| PointMaze-teleport   | 3/3  | 3/3 | 0/3  | 3/3   |
| AntMaze-large        | 3/3  | 2/3 | 0/3  | 1/3   |
| AntMaze-giant        | 3/3  | 3/3 | 0/3  | 3/3   |
| HumanoidMaze-large   | 3/3  | 0/3 | 0/3  | 0/3   |
| Cube-single          | 3/3  | 0/3 | 0/3  | 0/3   |
| Cube-double          | 3/3  | 0/3 | 0/3  | 0/3   |
| Cube-triple          | 3/3  | 0/3 | 0/3  | 0/3   |
| Puzzle-3x3           | 3/3  | 0/3 | 0/3  | 0/3   |
| Scene                | 3/3  | 0/3 | 0/3  | 0/3   |

retrieval and exposure. Table 14 reports slot retrieval top-1 by environment and age. Retrieval is well above the chance rate expected from the number of candidates but is not perfectly correlated with readout exposure: some manipulation decks have moderate retrieval yet strong full-memory exposure, because the exposed cue class is recoverable even when the exact mined patch is not the top match. This is consistent with our central claim that host-visible *exposure*, not internal retrieval, is the property that matters for a frozen interface.

**Table 14:** Slot-memory JEPA retrieval top-1 by environment and age (diagnostic only; not used for gating).

| Environment        | Age 4 | Age 8 | Age 15 |
|--------------------|-------|-------|--------|
| PointMaze-medium   | 0.340 | 0.370 | 0.219  |
| PointMaze-large    | 0.439 | 0.426 | 0.446  |
| PointMaze-giant    | 0.288 | 0.247 | 0.193  |
| PointMaze-teleport | 0.335 | 0.333 | 0.351  |
| AntMaze-large      | 0.444 | 0.433 | 0.431  |
| AntMaze-giant      | 0.266 | 0.223 | 0.190  |
| HumanoidMaze-large | 0.236 | 0.216 | 0.136  |
| Cube-single        | 0.762 | 0.816 | 0.684  |
| Cube-double        | 0.740 | 0.814 | 0.677  |
| Cube-triple        | 0.755 | 0.786 | 0.706  |
| Puzzle-3x3         | 0.918 | 0.900 | 0.771  |
| Scene              | 0.755 | 0.851 | 0.669  |

## E FULL NATIVE-USE TABLE

Table 15 reports per-environment-age executed success for the supported decks, with standard deviation over seeds and the recent-only, random-target, and oracle controls. The controller receives the *memory-selected* target rather than the true label, so recent-only and random targets cannot be repaired downstream; the gain column is full minus the stronger control. AntMaze-large/giant and HumanoidMaze-large are omitted because no audited local low-level controller exists for those bodies; they are capacity-only in the retention audit and are never counted as native-use successes.

## F MIXED-AGE GENERALIZATION AND THE AGE-ROUTER BOUNDARY

**Mixed-age result.** To test whether the interface is tied to the specific registered ages, we train the slot writer on a mixture of cue delays drawn from  $\{4, 8, 15\}$  and evaluate on the denser grid  $\{4, 6, 8, 10, 12, 15, 18\}$ ; the delays  $\{6, 10, 12, 18\}$  are never seen during training. Table 16 reports per-environment full-memory balanced accuracy across the evaluation grid. Every environment-age cell clears the same fail-closed gate as the fixed-age protocol, and full-memory exposure on the held-out delays is statistically indistinguishable from the trained delays while reset and no-state controls remain at chance. The delay at which evidence must be retained is therefore not memorized as a fixed offset: a single mixed-age interface is delay-robust across the tested range.

**Table 15:** Per-environment-age fixed-controller native use on supported decks. Full is mean  $\pm$  std executed success over seeds; gain is full minus the stronger of recent/random.

| Environment        | Age | Full              | Recent | Random | Oracle | Gain   |
|--------------------|-----|-------------------|--------|--------|--------|--------|
| PointMaze-medium   | 4   | 1.000 $\pm$ 0.000 | 0.244  | 0.244  | 1.000  | +0.756 |
| PointMaze-medium   | 8   | 1.000 $\pm$ 0.000 | 0.267  | 0.244  | 1.000  | +0.733 |
| PointMaze-medium   | 15  | 0.986 $\pm$ 0.017 | 0.194  | 0.244  | 1.000  | +0.742 |
| PointMaze-large    | 4   | 1.000 $\pm$ 0.000 | 0.256  | 0.244  | 1.000  | +0.744 |
| PointMaze-large    | 8   | 1.000 $\pm$ 0.000 | 0.244  | 0.244  | 1.000  | +0.756 |
| PointMaze-large    | 15  | 0.992 $\pm$ 0.008 | 0.225  | 0.244  | 1.000  | +0.747 |
| PointMaze-giant    | 4   | 1.000 $\pm$ 0.000 | 0.233  | 0.244  | 1.000  | +0.756 |
| PointMaze-giant    | 8   | 1.000 $\pm$ 0.000 | 0.261  | 0.244  | 1.000  | +0.739 |
| PointMaze-giant    | 15  | 0.961 $\pm$ 0.017 | 0.258  | 0.244  | 1.000  | +0.703 |
| PointMaze-teleport | 4   | 1.000 $\pm$ 0.000 | 0.483  | 0.556  | 1.000  | +0.444 |
| PointMaze-teleport | 8   | 1.000 $\pm$ 0.000 | 0.667  | 0.556  | 1.000  | +0.333 |
| PointMaze-teleport | 15  | 1.000 $\pm$ 0.000 | 0.678  | 0.556  | 1.000  | +0.322 |
| Cube-single        | 4   | 1.000 $\pm$ 0.000 | 0.219  | 0.244  | 1.000  | +0.756 |
| Cube-single        | 8   | 1.000 $\pm$ 0.000 | 0.225  | 0.244  | 1.000  | +0.756 |
| Cube-single        | 15  | 1.000 $\pm$ 0.000 | 0.250  | 0.244  | 1.000  | +0.750 |
| Cube-double        | 4   | 0.944 $\pm$ 0.029 | 0.281  | 0.281  | 0.944  | +0.664 |
| Cube-double        | 8   | 0.944 $\pm$ 0.029 | 0.236  | 0.281  | 0.944  | +0.664 |
| Cube-double        | 15  | 0.944 $\pm$ 0.029 | 0.233  | 0.281  | 0.944  | +0.664 |
| Cube-triple        | 4   | 0.911 $\pm$ 0.010 | 0.197  | 0.222  | 0.911  | +0.689 |
| Cube-triple        | 8   | 0.911 $\pm$ 0.010 | 0.222  | 0.222  | 0.911  | +0.689 |
| Cube-triple        | 15  | 0.911 $\pm$ 0.010 | 0.200  | 0.222  | 0.911  | +0.689 |
| Puzzle-3x3         | 4   | 1.000 $\pm$ 0.000 | 0.269  | 0.244  | 1.000  | +0.731 |
| Puzzle-3x3         | 8   | 0.956 $\pm$ 0.010 | 0.242  | 0.244  | 1.000  | +0.711 |
| Puzzle-3x3         | 15  | 0.828 $\pm$ 0.093 | 0.247  | 0.244  | 1.000  | +0.581 |
| Scene              | 4   | 0.881 $\pm$ 0.049 | 0.247  | 0.228  | 0.903  | +0.633 |
| Scene              | 8   | 0.842 $\pm$ 0.014 | 0.217  | 0.228  | 0.903  | +0.614 |
| Scene              | 15  | 0.839 $\pm$ 0.017 | 0.242  | 0.228  | 0.903  | +0.597 |

**Table 16:** Mixed-age training, per-environment full-memory balanced accuracy on the evaluation grid. Columns 6, 10, 12, and 18 are held out from training.

| Environment        | 4     | 6     | 8     | 10    | 12    | 15    | 18    |
|--------------------|-------|-------|-------|-------|-------|-------|-------|
| PointMaze-medium   | 1.000 | 1.000 | 1.000 | 0.998 | 1.000 | 0.991 | 0.982 |
| PointMaze-large    | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| PointMaze-giant    | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| PointMaze-teleport | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| AntMaze-large      | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| AntMaze-giant      | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| HumanoidMaze-large | 0.887 | 0.886 | 0.896 | 0.918 | 0.904 | 0.911 | 0.895 |
| Cube-single        | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Cube-double        | 0.998 | 0.998 | 0.998 | 0.998 | 0.998 | 0.998 | 0.998 |
| Cube-triple        | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Puzzle-3x3         | 0.973 | 0.981 | 0.983 | 0.981 | 0.986 | 0.978 | 0.957 |
| Scene              | 0.956 | 0.953 | 0.940 | 0.959 | 0.964 | 0.941 | 0.949 |

**What is still future work.** This mixed-age result demonstrates unseen-age *generalization*. It does not yet implement a learned explicit age router or an automatic maximum-age search. The next extension attaches a timestamp to each memory slot and trains a query network that scores stored slots by content match, timestamp, and confidence, then selects the top- $k$  slots; evaluation would report the maximum passing age found by a doubling-and-binary-search protocol. That answers a different question (how far back to retrieve, and how far back retrieval remains possible) than the one audited here (whether old evidence at a given delay is retained and exposed), and we therefore keep it out of the current claim.

## G CARRIER AND LOSS-COMPONENT ABLATIONS

We ablate the two design choices behind the mechanistic claim on the structured decks (Cube-single, Puzzle-3x3) at age 15. Table 17 varies the slot count:  $S=1$  is the single-vector carrier of the target-carrier grid, and accuracy rises with additional slots, confirming that the slot *set*, not a single compressed vector, carries the multi-part patch-set target. Table 18 removes individual loss terms

and swaps in the streaming writer; the full set objective and the streaming variant retain the exposure while dropping the contrastive or anti-collapse term degrades it.

**Table 17:** Slot-count (carrier) ablation on structured decks at age 15.

| Slots $S$         | Full  | Reset | No-state |
|-------------------|-------|-------|----------|
| 1 (single vector) | 0.726 | 0.201 | 0.227    |
| 2                 | 0.782 | 0.271 | 0.257    |
| 4                 | 0.838 | 0.255 | 0.229    |
| 8                 | 0.818 | 0.252 | 0.229    |
| 16                | 0.708 | 0.249 | 0.255    |

**Table 18:** Loss-component and streaming ablation on structured decks at age 15.

| Configuration           | Full  | Reset | No-state |
|-------------------------|-------|-------|----------|
| full loss (NCE+cos+std) | 0.818 | 0.252 | 0.229    |
| InfoNCE only            | 0.666 | 0.227 | 0.265    |
| no cosine               | 0.659 | 0.238 | 0.239    |
| no std reg.             | 0.812 | 0.257 | 0.211    |
| streaming ( $K=4$ )     | 0.671 | 0.237 | 0.242    |

## H ADDITIONAL ANALYSIS NOTES

**Why the controls are matched at the feature level.** An earlier and weaker design compares full memory against an unconditioned classifier. That comparison is too easy to pass, because class priors and endpoint pixels can carry substantial signal. Matching the reset and no-state controls to the *same* readout, endpoint, and held-out episodes forces any leaked signal to appear in all three conditions simultaneously, so a leaked deck cannot pass the gate in Eq. (1).

**Why a set target rather than a frame target.** A whole-frame latent target can be reconstructed from coarse scene statistics, which is sufficient for navigation but discards object-level structure. The mined patch-*set* target keeps several spatially distributed facts, so the memory must preserve more than a global summary. Empirically this is what separates the manipulation and scene decks as evidence ages (Tables 12, 13).

**Compute.** Each method sweep is  $12 \text{ decks} \times 3 \text{ ages} \times 3 \text{ seeds} = 108 \text{ cells}$ , run across three local GPUs. The four sweeps (slot, GRU, LSTM, Mamba-lite) reuse the same rendered caches, so the dominant cost is writer training rather than data generation. All figures and tables in this paper are regenerated in a single pass by `scripts/plot_paper_c.py`.